Antimicrobial Stewardship in Office Practice

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Introduction

- ☑ Antimicrobial stewardship (AMS) is defined as a coherent set of actions which promote the responsible use of antimicrobials.
- ☑ Most of the infections in outpatient practice are viral infections.
- ☑ However, antibiotic prescribing largely occurs in the outpatient setting and it is estimated that 50% of the time the antibiotics are inappropriate highlighting the need for stewardship in office practice.
- ☑ *Antimicrobial stewardship involves prescribing:*
  - The right **D**rug
  - In the right **D**ose
  - Through the right **D**elivery route
  - At the right **D**osing interval
  - For the right **D**uration
  - Right diagnosis is another "D" that needs to be aimed for along with the 5Ds of prescribing to avoid misdiagnosis or delayed diagnosis.
- ☑ A growing body of evidence demonstrates that antibiotic stewardship programs (AMSPs) reduce antibiotic overuse while improving patient outcomes.
- ☑ Antibiotic use in outpatient department (OPD) is mostly empiric. The empirical antibiotic choice should be made on basis of the following:
  - Suspected etiology
  - Site of infection
  - Host comorbid conditions including allergies and immunocompromising situation/organ dysfunction
  - Suspicion of drug resistance

- ☑ Appropriate cultures must be sent prior to starting antibiotics in bacteremia like enteric (blood culture) or urinary tract infections (urine culture).
- ☑ Point of care tests like rapid antigen test in acute pharyngitis, rapid malarial antigen test, dengue NS1 antigen, etc., is encouraged where they are clinically relevant.
- ☑ Initiation of antibiotics solely on basis of less reliable laboratory methods like Widal test is discouraged.
- ☑ Proper communication skills in outpatient settings are important for monitoring patients successfully during the course of illness.
- ☑ The core elements of antimicrobial stewardship practice in office settings are enumerated in **Table 1**.

TABLE 1: Core elements of antimicrobial stewardship practice in office settings.

Core elements	***Details***
Commitment	☑ Be committed every day ☑ May display posters of commitment but more important is to follow the same ☑ Display patient education posters such as "Be antibiotic Aware" "Avoid antibiotic abuse," etc.
Actions for policy and practice	☑ Follow evidence-based recommendations of IAP/ICMR or international guidelines ☑ Practice writing indication if writing antibiotic ☑ Set targets like not using antibiotics in clinically diagnosed viral URTI, improve counseling skills ☑ "Access" Category antibiotics encouraged ☑ Avoid antibiotics from "Watch and Reserve Categories" unless no other alternative ☑ AWaRe Classification available from https://www.who.int/publications/i/item/2021-aware-classification
Tracking	☑ Self-audit using antibiotic chart or electronic records ☑ Data regarding clinical outcome of patients who were not started antibiotics/requiring antibiotics in subsequent visits are helpful ☑ If in group practice, can have a peer audit
Education	☑ Stay updated and educate parents and colleagues regarding appropriate use of antibiotics ☑ Educate patients to not self-prescribe or buy over the counter medications, proper ways to reconstitute bottles, storage and completing course

(IAP: Indian Academy of Pediatrics; ICMR: Indian Council of Medical Research; URTI: upper respiratory tract infection)

Source: Centers for Disease Control and Prevention. (2021). CDC Core elements for antimicrobial stewardship in outpatient practices. [Online] Available from https://www.cdc.gov/antibiotic-use/core-elements/outpatient.html#:~:text=The%20four%20core%20elements%20of,reporting%2C%20and%20education%20and%20expertise [Last accessed December, 2022].

The common clinical scenarios in OPD practice where systemic antibiotic use can be easily avoided are enumerated in **Table 2**.

TABLE 2: Common clinical scenarios in OPD practice where systemic antibiotic use can be easily avoided.

Clinical scenario	***Clinical signs and symptoms***	***Parent counseling***
Fever	☑ Child active and playful in between fever, good response to paracetamol, generalized symptoms such as cold, cough, body ache, and loose motions ☑ Viral fevers generally start spacing out after 72 hours	To follow-up if fever intensity/frequency worsens or child becomes sick looking in between fever
Throat pain	Associated with cough, absence of exudates on examination, and presence of ulcers	☑ To follow-up if throat pain worsens after 48 hours, look for any new development of exudates or tender cervical LNs ☑ In such a situation, rapid throat antigen test might be useful
Earache	Unilateral, <48 hours, child above 2 years, no systemic toxicity	To follow-up if earache persists
Cold	Short-duration, watery/blocked nose, no tenderness over sinuses, and nonsick looking	Saline nose drops are helpful
Cough	Associated with generalized symptoms of fever, cold, nonsick looking, and wheeze. No localized crepitations	☑ Counsel regarding cough management ☑ X-ray chest if done for suspected LRTI, hyperinflated fields with few infiltrates favor a viral etiology
Diarrhea	Watery loose motions without visible blood	☑ Focus on maintaining hydration ☑ No need for stool test if no visible blood
Skin infections	Single impetigo, small pustules or boils where local creams will suffice	
Abdominal pain	Soft abdomen without localized tenderness/guarding/rigidity, not associated with blood in stools, no systemic toxicity	Check for constipation and manage appropriately
Pain during micturition	No fever/systemic toxicity and presence of rash over vulval region	Follow-up after prescribing skin protective cream/do urine routine to rule out pyuria

(LN: lymph nodes; LRTI: lower respiratory tract infection)

The common clinical scenarios where antibiotics are used are enumerated in **Table 3**.

TABLE 3: The common clinical scenarios where antibiotics are used.

Clinical scenario	***Clinical signs and symptoms***	***Management***
Fever	Child is sick/lethargic in between fever and poor response to paracetamol	☑ Investigate to find the etiology based on associated symptoms ☑ CBC and CRP helps in screening ☑ Urine routine in cases of fever without focus ☑ Blood culture must be sent if clinical suspicion of enteric fever ☑ As per clinical clues, investigate for malaria, dengue, Rickettsia, and leptospirosis ☑ X-ray of chest helps with associated respiratory symptoms or PUO ☑ USG abdomen for localized findings in abdomen or PUO
Throat pain	Absence of cough, presence of exudates on examination, and tender jugular LNs	In case systemic examination reveals splenomegaly, do CBC to look for lymphocytosis as in infectious mononucleosis
Earache	Bilateral, >48 hours, systemic toxicity, TM congested and bulging	Start antibiotics and follow-up otoscopy for improvement and at end of treatment
Diarrhea	Visible blood in stools (dysentery)/rice water stools as in cholera and loose motions in a sick looking child	☑ Start antibiotics ☑ Stool microscopy/PCR tests if suspected cholera.
Cough	Associated with localized symptoms of fever, sick looking, and localized crepitations	For children above 5 years, consider macrolides for atypical pneumonia
Skin infections	Multiple impetigo or abscess/cellulitis	Drain purulent collections
Abdominal pain	Soft abdomen with local tenderness/ guarding	Rule out surgical emergencies such as appendicitis, disproportionate tachycardia, and board-like rigidity are red flag signs of danger
Pain during micturition	☑ Absence of local rash ☑ Presence of fever	☑ Urine routine to look for pyuria ☑ If pyuria confirmed, send urine culture before starting antibiotics and modify based on the reports

(CBC: complete blood count; CRP: C-reactive protein; LN: lymph node; PCR: polymerase chain reaction; PUO: pyrexia of unknown origin; TM: tympanic membrane; USG: ultrasonography)

The commonly used antibiotics in office practice are enumerated in **Table 4**.

TABLE 4: The commonly used antibiotics in office practice.

Antibiotic	***Class and mechanism of action***	***Spectrum***	***Recommended use***	***Oral dose***
Amoxicillin	Aminopenicillin β-lactam Inhibits cell wall synthesis	*Gram-positive:* *Streptococcus spp.*, *Enterococcus spp.*, and *Listeria monocytogenes* *Gram-negative:* *Haemophilus influenzae*, *Escherichia coli*, *Proteus mirabilis*, *Salmonella*, and *Shigella spp.*	Streptococcal pharyngitis/ tonsillitis (first line)	50 mg/kg/dose 12 hourly (maximum 1,000 mg/day) after food
			Pneumonia/acute sinusitis (first line)	45 mg/kg/day in three divided doses
			Otitis media (first line)	45 mg/kg/day for 10 days, in three divided dose after food In case of failure, high dose amoxicillin with 40–50 mg/kg/dose 12 hourly
			IE prophylaxis before dental procedure	50 mg/kg ½–1 hour before procedure
Co-amoxiclav	Aminopenicillin plus β-lactamase inhibitor Amoxicillin inhibits cell wall synthesis while clavunate inhibits certain β-lactamases	*Gram-positive:* *Staphylococcus aureus* *Streptococcus* *Gram-negative:* B-lactamase producing *H. influenzae*, *Moraxella catarrhalis*, *β-lactamase producing E. coli*, and anaerobes *Bacteroides* species, including *Bacteroides fragilis* (β-lactamase and non-β-lactamase-producing) *Fusobacterium species* (β-lactamase and non-β-lactamase-producing) *Peptostreptococcus*	Pneumonia Sinusitis (not responding to amoxicillin)	45 mg/kg/day of amoxicillin in two divided doses
			Otitis media (not responding to amoxicillin/ suspicion of resistance)	40–50 mg/kg/dose of amoxicillin in two divided doses (after food)
			Cellulitis (second line, first generation cephalosporins preferred as Co-amoxiclav has a broader activity)	45 mg/kg/day of amoxicillin in two divided doses
			UTI	30-40 mg/kg/day of amoxicillin in two divided doses

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Antibiotic	***Class and mechanism of action***	***Spectrum***	***Recommended use***	***Oral dose***
Cloxacillin	Semisynthetic penicillin Inhibits cell wall synthesis	*Staphylococcus aureus* *Streptococcus*	First line for MSSA infections including osteomyelitis	25–50 mg/kg/day in four divided dose
Cefadroxil	First generation cephalosporin Inhibits cell wall synthesis	*Gram-positive:* Streptococci, *Staphylococcus aureus* *Gram-negative: Moraxella catarrhalis, Escherichia coli, Klebsiella pneumoniae, Proteus mirabilis*	Skin and soft tissue infections (first line)	30 mg/kg/day in two divided dose (maximum 2 g/day)
			Group A streptococcal pharyngitis (in penicillin allergy, avoid in immediate hypersensitivity to penicillin)	30 mg/kg once daily (maximum = 1 g)
Cephalexin	First generation cephalosporin Inhibits cell wall synthesis	*Gram-positive: Streptococci, Staphylococcus aureus* *Gram-negative:* Minimal activity against *E. coli, Proteus,* and *Klebsiella*	Skin and soft tissue infections (first line)	(25–50 mg/kg/day) in 3–4 divided doses
			Pneumonia/otitis media	75–100 mg/kg/day in four divided doses in severe infections
			Group A streptococcal pharyngitis (in penicillin allergy, avoid in immediate hypersensitivity to penicillin)	20 mg/kg/dose (maximum 500 mg/dose) 12 hourly for 10 days
			UTI	50–70 mg/kg/day in 2–3 divided doses
			UTI prophylaxis	10 mg/kg/day
			IE prophylaxis (in penicillin allergy, avoid in immediate hypersensitivity to penicillin)	50 mg/kg single dose half an hour before procedure

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Antibiotic	Class and mechanism of action	Spectrum	Recommended use	Oral dose
Cefixime	Third generation cephalosporin Inhibits cell wall synthesis	Gram-positive (minimal activity-no clinical use) *Streptococcus pneumoniae,* and *Staphylococcus* *Gram-negative: E. coli, Salmonella, H. influenzae, Moraxella catarrhalis, Neisseria meningitidis, Neisseria gonorrhoeae, Enterobacteriaceae,* and *E. coli*	UTI and bacillary dysentery (first line)	10 mg/kg/day
			Enteric fever (first line)	20 mg/kg/day, maximum dose 1,200 mg/day
Cefpodoxime	Third generation cephalosporin Inhibits cell wall synthesis	*Gram-positive: S. pneumoniae, Streptococcus pyogenes*, and *Staphylococcus aureus* *Gram-negative: H. influenzae, Moraxella catarrhalis, E. coli*, and *Klebsiella* Not effective against *Pseudomonas aeruginosa, Enterococcus, and Bacteroides fragilis*	*Mainly in respiratory infections:* Pneumonia, otitis media, sinusitis, and pharyngotonsillitis	10 mg/kg/day in two divided dose
Azithromycin	Macrolide Protein synthesis inhibitor (50S ribosome)	*Atypical bacteria Mycoplasma, Chlamydia*, and *Legionella* *Gram-negative Salmonella, Shigella, H. influenzae*, and *Moraxella catarrhalis* *Gram-positive Streptococcus pyogenes* *Mycobacterium avium complex*	Second line for enteric (preferable to reserve for cefixime-resistant enteric fever) *Mycoplasma* infections (first line) Dysentery (second line) In penicillin allergy	20 mg/kg/day in two divided dose (maximum 1 g/day) 10 mg/kg/day
Clindamycin	Macrolide Inhibits protein synthesis (50S ribosome)	Anaerobes such as *B. fragilis, Peptostreptococcus, Fusobacterium, Staphylococcus* (including MRSA) In addition, has an antitoxin effect	CA-MRSA infections, anaerobic infections especially of head, neck and face, and lungs	30–40 mg/kg/day in 3–4 divided doses
			IE prophylaxis (in penicillin allergy)	20 mg/kg ½–1 hour before procedure

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Antibiotic	Class and mechanism of action	Spectrum	Recommended use	Oral dose
Metro-nidazole	Nitroimidazole Inhibits nucleic acid synthesis	Only active against anaerobes such as *Bacteroides and Entamoeba histolytica*	Amoebic dysentery, amoebic liver abscess, giardiasis, and *H. pylori* infection	30–40 mg/kg/day in 3–4 divided doses
Cotri-moxazole	Trimethoprim-sulfametho-xazole inhibits folate synthesis	Broad-spectrum includes gram-negative such as *E. coli, Proteus, Klebsiella, Vibrio cholera,* and *H. influenzae,* Gram-positive such as *Streptococcus pneumoniae, Staphylo-coccus, and Nocardia*	*Not used in age <3 months/G6PD deficiency:* UTI Shigellosis *Brucella* Cholera	8–12 mg/kg/day, dosing based on trimethoprim component
			UTI prophylaxis	1–2 mg/kg/day
Doxycycline	Tetracycline Protein synthesis inhibitor	Broad-spectrum, includes many gram-positive and gram-negative including *Vibrio cholera, Brucella, Anthrax, Rickettsia, Mycoplasma, Chlamydia, Plasmodium falciparum*, etc.	Rickettsial infections and cholera *Brucella* CA-MRSA infections. Macrolide-resistant *Mycoplasma pneumoniae*	4 mg/kg/day in two divided doses (maximum 200 mg/day)

(CA-MRSA: community-associated MRSA; G6PD: glucose-6-phosphate dehydrogenase; IE: infective endocarditis; MRSA: methicillin-resistant *Staphylococcus aureus*; MSSA: methicillin-susceptible *Staphylococcus aureus*; UTI: urinary tract infection)

Further Reading

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